# Machine Learning 2026 • Tutorial 1

Mathematical Readiness and A0 Clinic Name: _____

07 August 2026 • B2, 08:30–09:50 • Jagdish/Harsha Patel Learning Theatres

Everything here is plain mathematics. Nothing needs machine learning today — and every piece returns later wearing a machine-learning name. **No calculator is needed:** every value you cannot do in your head is given to you. Work in the boxes; bring one unresolved question to Tutorial 2.

## Station 1 Shapes: when does a product exist?

A product $\mathbf{AB}$ with $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{B} \in \mathbb{R}^{p\times q}$ exists only if $n = p$, and then $\mathbf{AB} \in \mathbb{R}^{m\times q}$. *Inner dimensions agree; outer dimensions survive.* A column vector $\mathbf{v} \in \mathbb{R}^d$ is a $d \times 1$ matrix, so $\mathbf{u}^\top\mathbf{v}$ is a scalar while $\mathbf{uv}^\top$ is $d \times d$.

**1.1** For $\mathbf{A} \in \mathbb{R}^{3\times 2}$ and $\mathbf{B} \in \mathbb{R}^{2\times 4}$: which of $\mathbf{AB}$, $\mathbf{BA}$ exists, and with what shape?

**1.2** Let $\mathbf{X} \in \mathbb{R}^{n\times d}$, $\theta \in \mathbb{R}^d$, $\mathbf{y} \in \mathbb{R}^n$. Fill in the dimensions.

| Expression | Dimension | Meaning |
|---|---|---|
| $\mathbf{X}\theta$ | _____ | one prediction per observation |
| $\mathbf{y} - \mathbf{X}\theta$ | _____ | residual (error) vector |
| $\mathbf{X}^\top\mathbf{y}$ | _____ | feature–target aggregate |
| $\mathbf{X}^\top\mathbf{X}$ | _____ | feature–feature aggregate |

Which of $\mathbf{Xy}$, $\mathbf{X}^\top\mathbf{y}$, $\theta^\top\mathbf{X}^\top\mathbf{y}$ is invalid, and why?

**1.3** Let $\mathbf{x} \in \mathbb{R}^4$, $\mathbf{W}_1 \in \mathbb{R}^{6\times 4}$, $\mathbf{b}_1 \in \mathbb{R}^6$, $\mathbf{W}_2 \in \mathbb{R}^{2\times 6}$, $\mathbf{b}_2 \in \mathbb{R}^2$. Give the shape of every intermediate object in $\mathbf{z} = \mathbf{W}_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$. Could you multiply in the other order?

***Later in this course:*** *this chain is the* forward pass of a two-layer network *— four inputs, six hidden units, two outputs.*

## Station 2 Two functions, studied as functions

**2.1** Let $\sigma(z) = \dfrac{1}{1+e^{-z}}$. Given $e^{-2} \approx 0.135$ and $e^2 \approx 7.39$, compute $\sigma(0)$, $\sigma(2)$, $\sigma(-2)$; state both limits as $z \to \pm\infty$; and check $\sigma(-z) = 1 - \sigma(z)$ on your numbers.

***Later in this course:*** *$\sigma$ squeezes $\mathbb{R}$ into $(0,1)$ and returns as the function that turns a score into a probability.*

**2.2** Let $m(z) = \max(0, z)$. Sketch it (axes below); compute $m(-3)$, $m(0)$, $m(2)$; give $m'(z)$ for $z > 0$ and $z < 0$; then say what $m'(0)$ is — be honest.

***Later in this course:*** *this is ReLU, the most used activation in deep networks — and that corner will matter.*

## Station 3 Finding extrema, and checking them

**The test.** If $f$ is twice differentiable and $f'(u^\star) = 0$, then $f''(u^\star) > 0$ gives a local minimum, $f''(u^\star) < 0$ a local maximum, and $f''(u^\star) = 0$ is inconclusive.

**3.1** For $h(w) = (w-3)^2 + 2$: find the critical point, apply the test, and state the minimum value.

**3.2** For $g(u) = u^4 - 2u^2$: find *every* critical point, classify each with the test, give $g$ at each, and say how many *global* minima there are.

***Later in this course:*** *"one valley" is called* convexity*. Objectives with many valleys are why optimization in machine learning is genuinely hard. The derivations come later — the test you can already run.*

## Station 4 A Bayes calculation, start to finish

Floods occur on 10% of monsoon days. Rainfall behaves differently on the two kinds of day:

$$\text{flood: } x \sim \mathcal{N}(120, 30^2) \text{ mm}, \qquad \text{no flood: } x \sim \mathcal{N}(40, 20^2) \text{ mm}, \qquad p(x) = \frac{c}{s}\, e^{-(x-m)^2/(2s^2)}.$$

Today it rained $x = 90$ mm. *You will never need the constant $c$* — watch it cancel. Values you may need, all given: $e^{2.625} \approx 13.8$, $e^{-1.5} \approx 0.223$, $e^{8} \approx 2981$.

**4.1** Write the exponent $-\dfrac{(x-m)^2}{2s^2}$ at $x = 90$ under each hypothesis. (Both are clean fractions.)

**4.2** Form the ratio $\dfrac{p(90 \mid \text{flood})}{p(90 \mid \text{no flood})}$. Show that $c$ cancels, that the standard deviations leave $\frac{20}{30}$, and that the exponents combine into a single power of $e$. Evaluate it.

**4.3** The prior odds of flood against no flood are $0.10 : 0.90$, i.e. $1 : 9$. Multiply prior odds by the ratio from 4.2 to get the posterior odds, then convert odds to a probability using $p = \dfrac{\text{odds}}{1 + \text{odds}}$.

**4.4 (homework)** Repeat 4.1–4.3 at $x = 60$ and $x = 120$ mm.

***Later in this course:*** *this is exactly how a* generative *model predicts — model how each kind of day produces data, then invert with Bayes' rule. The odds form you used is the same rule, rearranged so the constants never appear.*

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**Where today points.** A0 (Mathematical Readiness) is open on the portal, due **14 August, 23:59 IST**. Shapes → vectors, matrices and dimension checks. Functions → reading and evaluating functions. Extrema → derivatives and optimization. Bayes → probability and conditioning.

**Response standard.** For every expression you write: name the object type, state its dimension, and state the assumption behind the next step.

**Working space** use for 4.4, or for anything you want to redo